

HR Employee Attrition ❖❖❖ Logistic Regression Project Report

Logistic Regression — Predicting Employee Attrition (HR Analytics)

A binary classification project predicting whether an employee will leave the company (`left`), using satisfaction level, average monthly hours, and recent promotion history — built with `scikit-learn` 's Logistic Regression on a real-world HR dataset of ~15,000 employees.

1. Project Objective

Identify which employee attributes have a clear, measurable impact on attrition, and build a logistic regression model that predicts whether an employee will leave the company using the features narrowed down through exploratory analysis.

2. Dataset

Attribute	Detail
File	<code>HR_comma_sep.csv</code>
Observations	14,999 employees
Missing values	None
Target variable	<code>left</code> (1 = left the company, 0 = stayed)
Attrition rate	23.8% (3,571 of 14,999 employees left)

Available columns: `satisfaction_level` , `last_evaluation` , `number_project` , `average_monthly_hours` , `time_spend_company` , `Work_accident` , `left` , `promotion_last_5years` , `Department` , `salary` .

Features used in the model:

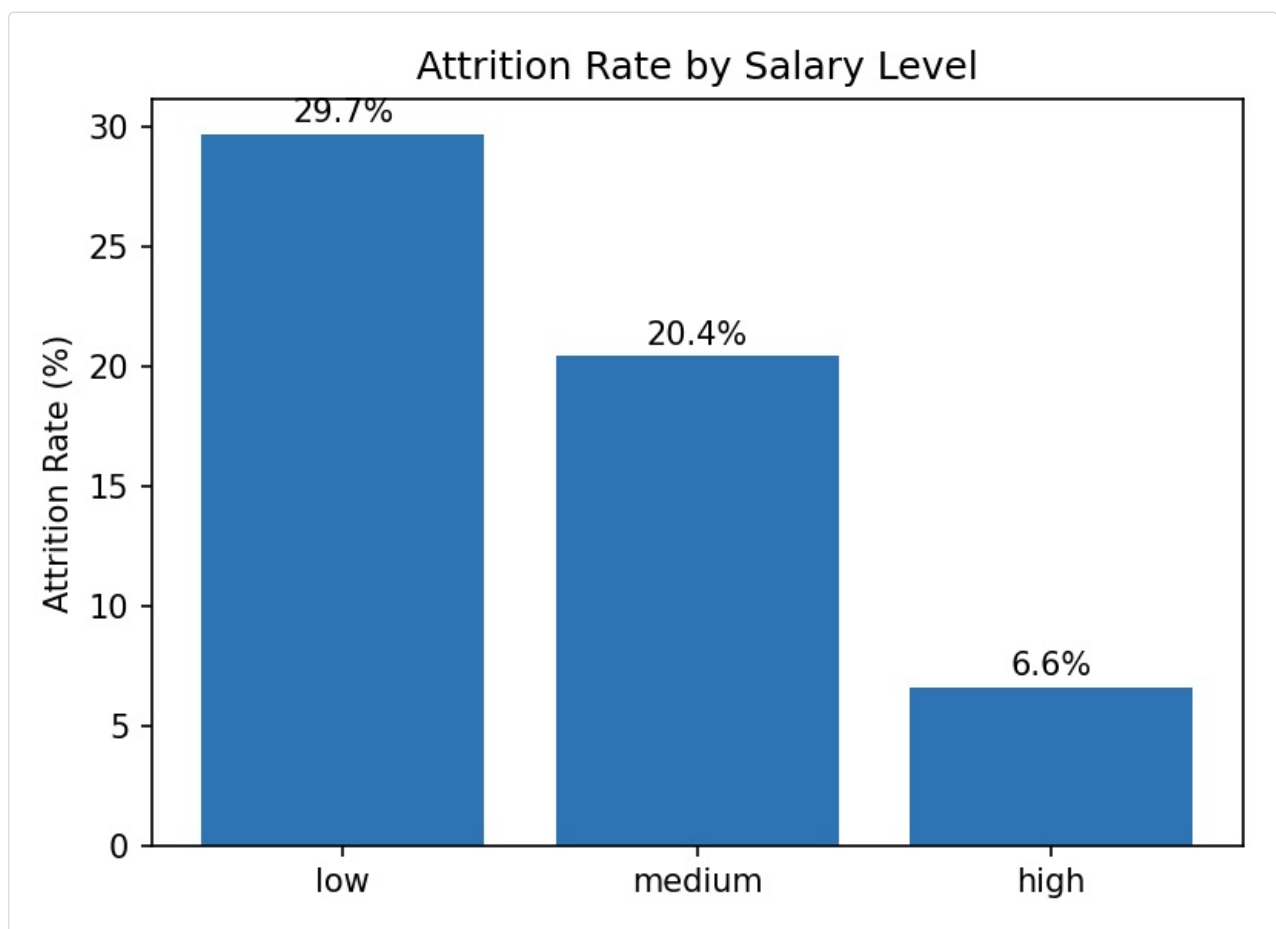
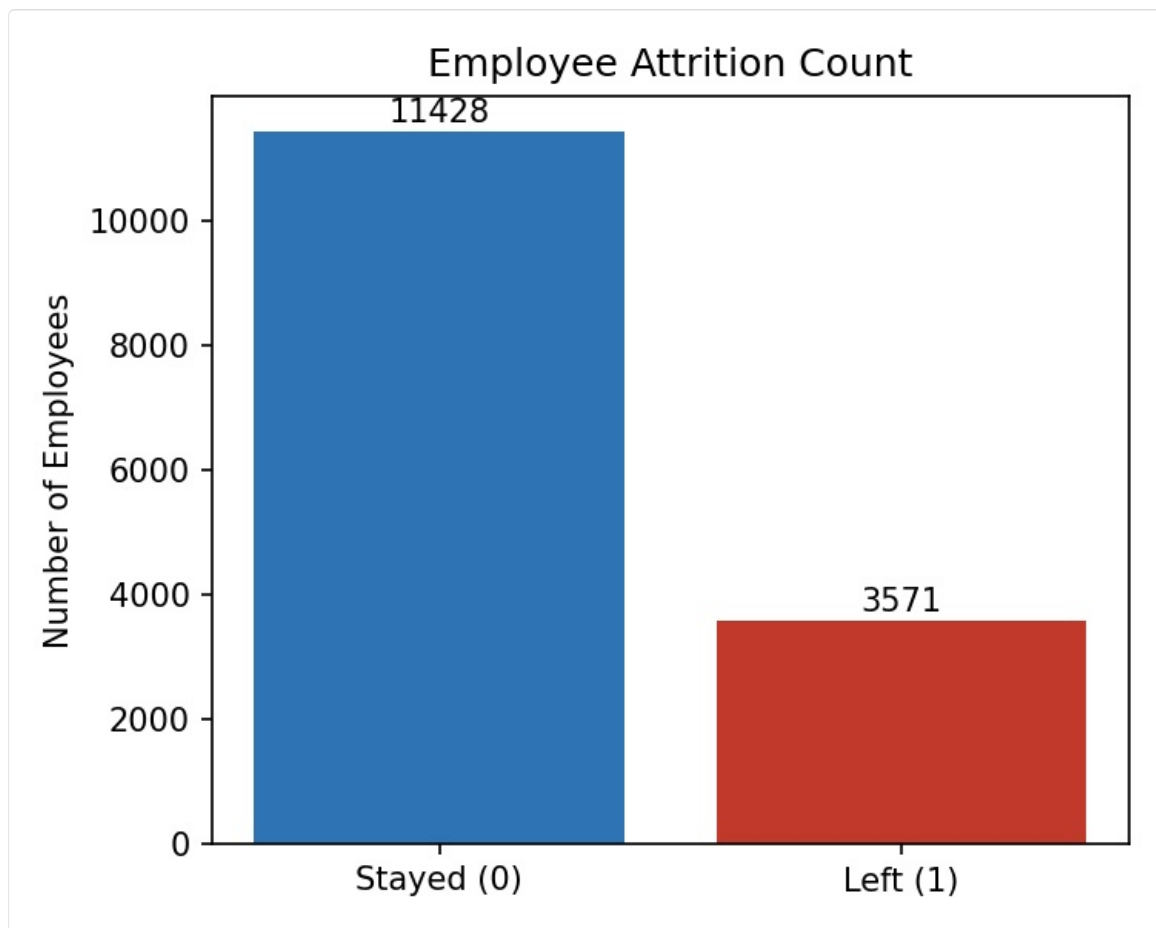
Feature	Description
<code>satisfaction_level</code>	Employee's self-reported job satisfaction (0-1)
<code>average_monthly_hours</code>	Average hours worked per month
<code>promotion_last_5years</code>	Whether the employee was promoted in the last 5 years (1/0)

3. Tools & Libraries

- **Python 3**
- `pandas` — data handling
- `matplotlib` , `seaborn` — exploratory visualization
- `scikit-learn` — train/test split, `LogisticRegression` , evaluation metrics

4. Exploratory Data Analysis

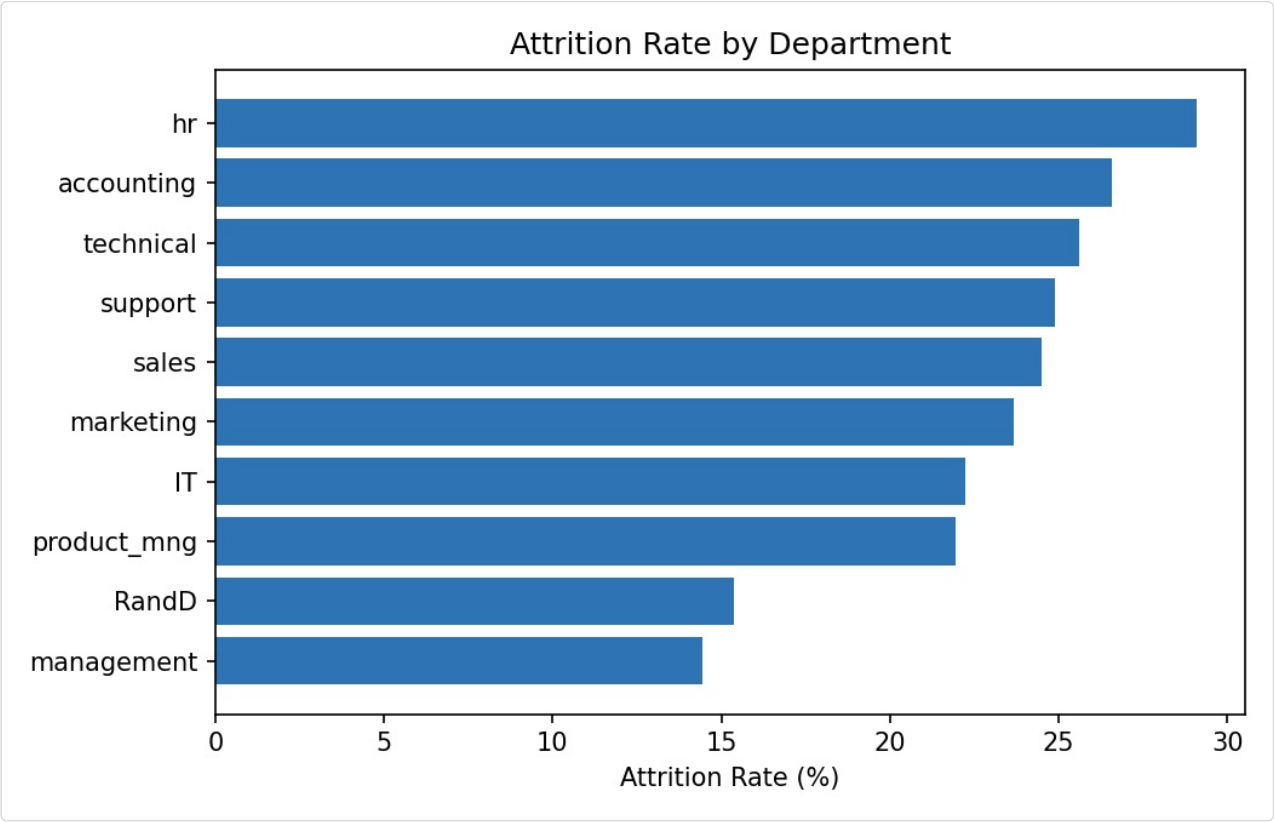
Before modeling, the impact of salary and department on attrition was examined, since these are the two most actionable levers for HR policy:



Salary Level		Attrition Rate	
Low		29.7%	

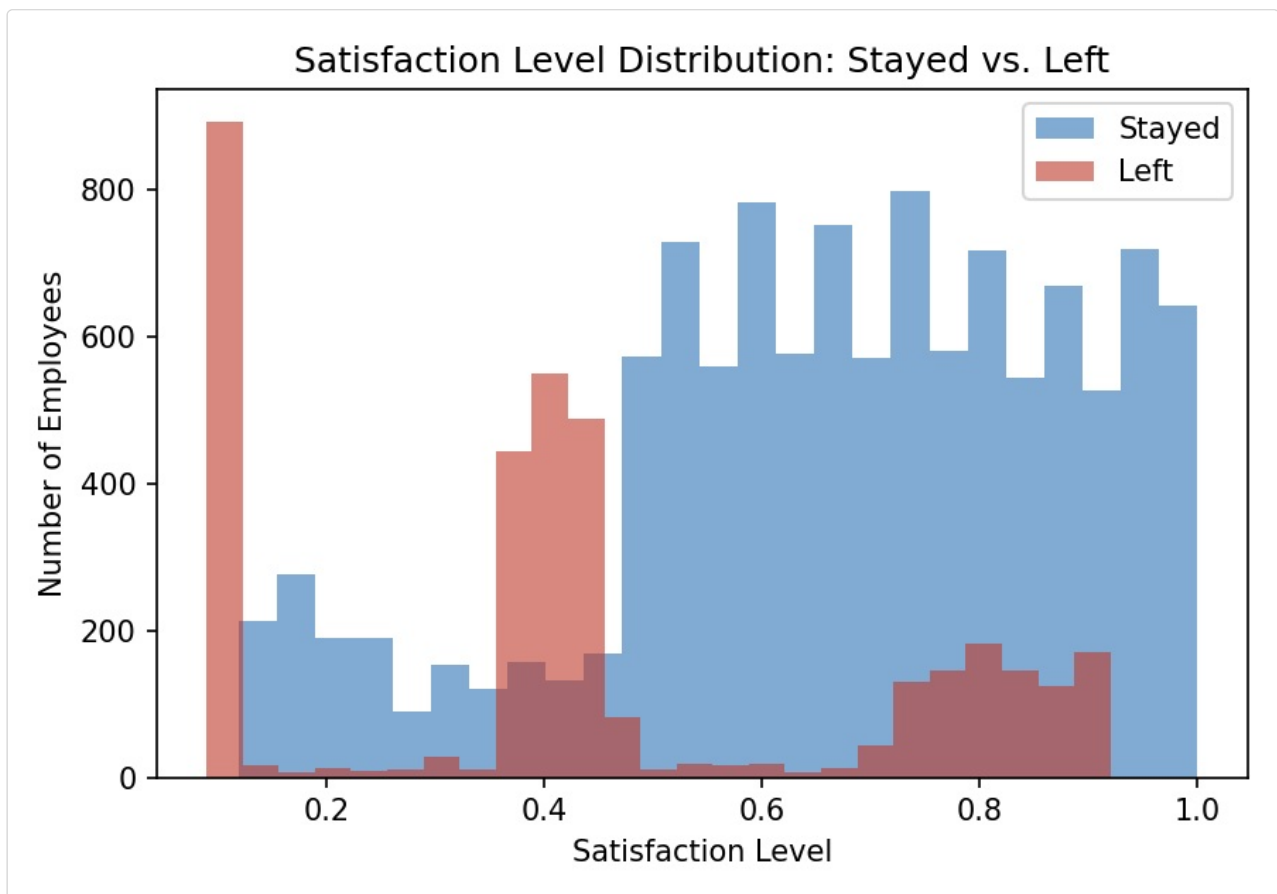
Medium	20.4%
High	6.6%

Salary has a strong, clear relationship with attrition — employees on low salaries leave at more than **4x the rate** of employees on high salaries.



Department	Attrition Rate
HR	29.1%
Accounting	26.6%
Technical	25.6%
Support	24.9%
Sales	24.5%
Marketing	23.7%
IT	22.2%
Product Mgmt	22.0%
R&D	15.4%
Management	14.4%

Department shows a **milder** effect than salary — a roughly 2x spread between the highest (HR) and lowest (Management) attrition rates, compared to salary's 4.5x spread.



Interestingly, employees who left do **not** simply have uniformly low satisfaction — the distribution is bimodal, with one cluster of clearly dissatisfied leavers (satisfaction < 0.2) and a second, smaller cluster of moderately satisfied leavers (satisfaction ~0.7–0.9), who likely left for reasons such as overwork or lack of promotion rather than dissatisfaction itself.

Feature correlation with attrition (`left`):

Feature	Correlation with <code>left</code>
<code>satisfaction_level</code>	-0.388 (strongest)
<code>time_spend_company</code>	+0.145
<code>Work_accident</code>	-0.155
<code>average_monthly_hours</code>	+0.071
<code>number_project</code>	+0.024
<code>last_evaluation</code>	+0.007
<code>promotion_last_5years</code>	-0.062

`satisfaction_level` is by far the most correlated numeric feature with attrition, which is why it was selected as the primary driver for the model below, alongside `average_monthly_hours` (a proxy for overwork) and `promotion_last_5years` (a proxy for career growth).

5. Methodology

1. **Load and inspect the dataset** — confirmed 14,999 rows, no missing values, and a 23.8% attrition rate.
2. **Exploratory analysis** — examined attrition by salary and department (above) to identify which variables have a clear, direct relationship with attrition before selecting model inputs.
3. **Feature and target selection** — chose `satisfaction_level`, `average_monthly_hours`, and `promotion_last_5years` as the model's inputs, with `left` as the target.
4. **Train/test split** — 80% train / 20% test (`random_state=100`), giving 11,999 training and 3,000 test employees.
5. **Model training** — `sklearn.linear_model.LogisticRegression` fit on the three selected features.

6. **Evaluation** — accuracy, confusion matrix, and classification report on the held-out test set.

```
X = df[['satisfaction_level', 'average_monthly_hours', 'promotion_last_5years']]
y = df['left']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=100)

model = LogisticRegression()
model.fit(X_train, y_train)
```

6. Model Results

Term	Coefficient
satisfaction_level	-3.667
average_monthly_hours	+0.0024
promotion_last_5years	-1.529
Intercept	+0.406

- satisfaction_level has, by far, the largest effect — each 1-point increase in satisfaction (on the 0-1 scale) sharply reduces the log-odds of leaving. This matches the strong correlation found during EDA.
- promotion_last_5years also has a large negative coefficient — being promoted in the last 5 years substantially reduces the odds of leaving.
- average_monthly_hours has a very small positive coefficient on its raw (unscaled) scale — its effect is real but modest per additional hour worked.

7. Model Evaluation

Metric	Value
Train / Test split	80% / 20% (11,999 / 3,000 employees)
Accuracy	77.4%

Confusion Matrix (test set, n=3,000)

	Predicted: Stayed	Predicted: Left
Actual: Stayed	2,112 (TN)	138 (FP)
Actual: Left	540 (FN)	210 (TP)

Classification Report

Class	Precision	Recall	F1-score	Support
0 — Stayed	0.80	0.94	0.86	2,250
1 — Left	0.60	0.28	0.38	750
Accuracy			0.77	3,000
Macro avg	0.70	0.61	0.62	3,000
Weighted avg	0.75	0.77	0.74	3,000

- The model is **much better at identifying employees who will stay (94% recall)** than employees who will leave (28% recall).
- Given the 23.8% baseline attrition rate, a model that always predicted "stayed" would already score ~76.2% accuracy — meaning the model's 77.4% accuracy is only a marginal improvement in raw accuracy terms, even though it does add real signal (visible in the non-trivial precision/recall on the "left" class).

- This is a common and important pattern in imbalanced HR/churn datasets: **accuracy alone is a misleading headline metric**; recall on the minority (attrition) class is the number that actually matters for a retention program.

Example prediction

```
sample_employee = [[0.4, 250, 0]] # satisfaction=0.4, avg monthly hours=250, no recent promotion
model.predict(sample_employee)    # → 0 (predicted to stay)
model.predict_proba(sample_employee) # → [0.61 stay, 0.39 leave]
```

An employee with below-average satisfaction (0.4), a high monthly workload (250 hours), and no recent promotion is predicted to stay, but with meaningfully elevated leave-risk (39%) — illustrating how the model outputs a graded risk score rather than just a hard yes/no, which is more useful in practice for prioritizing retention outreach.

8. Key Insights

- **Low satisfaction is the single strongest driver of attrition** in this dataset, both in raw correlation and in the fitted model's coefficients.
- **Salary tier has a very strong relationship with attrition** (29.7% for low salary vs. 6.6% for high salary) — a clear, actionable lever for HR to prioritize retention budget.
- **Lack of promotion compounds the risk** — the model's negative coefficient on `promotion_last_5years` confirms career stagnation is a meaningful attrition driver independent of satisfaction.
- **The satisfaction-vs-attrition relationship is not purely linear** — a distinct group of moderately-satisfied employees also leave, suggesting overwork or blocked career growth as a separate attrition pathway worth investigating with additional features (`number_project` , `time_spend_company`).
- **The model catches only 28% of employees who actually leave** — on its own it would not be reliable for a real early-warning system without further tuning.

9. Limitations & Next Steps

- **Class imbalance (23.8% positive)** likely explains the low recall on the "left" class; class weighting (`class_weight='balanced'`), oversampling (SMOTE), or threshold tuning would likely improve minority-class detection.
- **Only 3 of 9 available features were used** — `salary` and `Department` showed a clear relationship with attrition during EDA but were not included in the model as numeric/encoded features; one-hot encoding these categorical variables and adding them to the feature set is a natural next step.
- **No regularization tuning or cross-validation** was performed; a `GridSearchCV` over `C` (and potentially comparing to a tree-based model like Random Forest) would likely improve both accuracy and recall.
- **Correlation ≠ causation** — the relationships identified here (e.g., low satisfaction → attrition) are associative; a causal HR intervention study would need a controlled comparison.

10. How to Run

```
pip install pandas matplotlib seaborn scikit-learn
jupyter notebook logreg_practice_ex2_Answer.ipynb
```

11. Repository Contents

File	Description
<code>logreg_practice_ex2_Answer.ipynb</code>	Main analysis notebook
<code>HR_comma_sep.csv</code>	Source dataset
<code>attrition_count.png</code>	Overall attrition class balance

<code>salary_vs_retention.png</code>	Attrition rate by salary tier
<code>department_vs_retention.png</code>	Attrition rate by department
<code>satisfaction_distribution.png</code>	Satisfaction level distribution, stayed vs. left
<code>README.md</code>	This report

12. Conclusion

This project demonstrates a full HR analytics workflow: exploring which factors relate to employee attrition (salary and department show clear, strong effects), selecting features based on that analysis, and fitting a logistic regression model that achieves 77.4% accuracy overall. The key finding — that low satisfaction and lack of promotion are the dominant drivers of attrition — provides a clear, actionable narrative for HR stakeholders, while the model's low recall on actual leavers (28%) highlights a concrete next step: prioritizing minority-class detection over raw accuracy for any real retention-risk tool built from this data.

Tech stack: Python · pandas · matplotlib · seaborn · scikit-learn